

Risk-weighted apoB outperforms traditional lipid biomarkers in predicting residual cardiovascular risk in statin-treated hypertriglyceridemic patients: post hoc analysis of the REDUCE-IT placebo arm

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Concept

Risk-weighted apoB (RW-apoB) is a simple, integrated metric of the atherogenic risk of apoB-containing lipoproteins (Rehman MB & Björnson E et al. EHJ 2026)

On a per particle basis triglyceride-rich lipoproteins (TRL) are 4 to 5 times, and lipoprotein(a) (Lp(a)) 6 to 7 times more atherogenic than LDL.

To determine RW-apoB, the proportion of apoB in TRL, Lp(a), and LDL is estimated from standard lipid profiles and each component weighted according to its relative atherogenicity (Panel A). In multiple cohorts, RW-apoB was a more accurate risk marker than LDL-C, non-HDL-C or apoB. (Panel B)

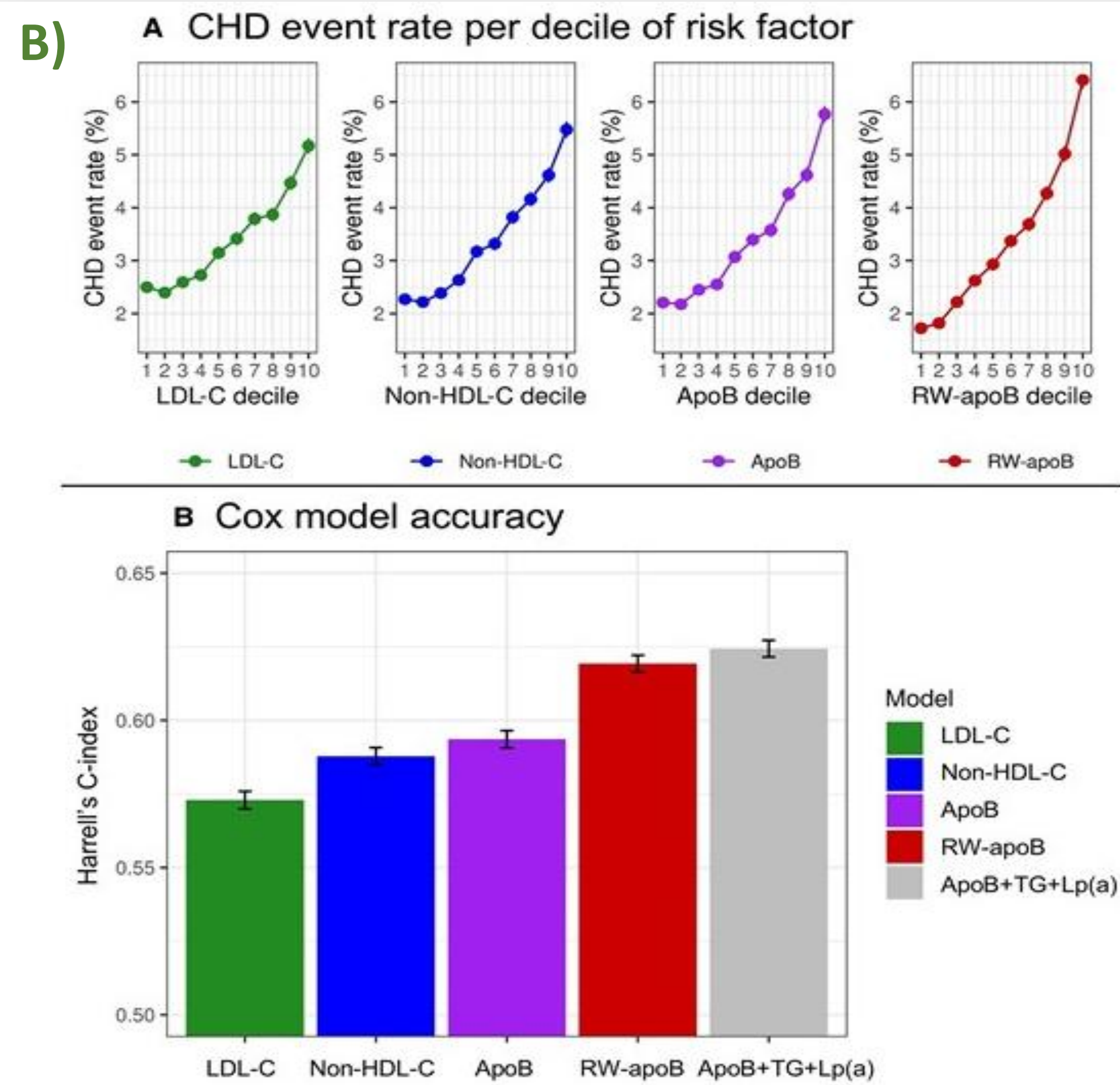
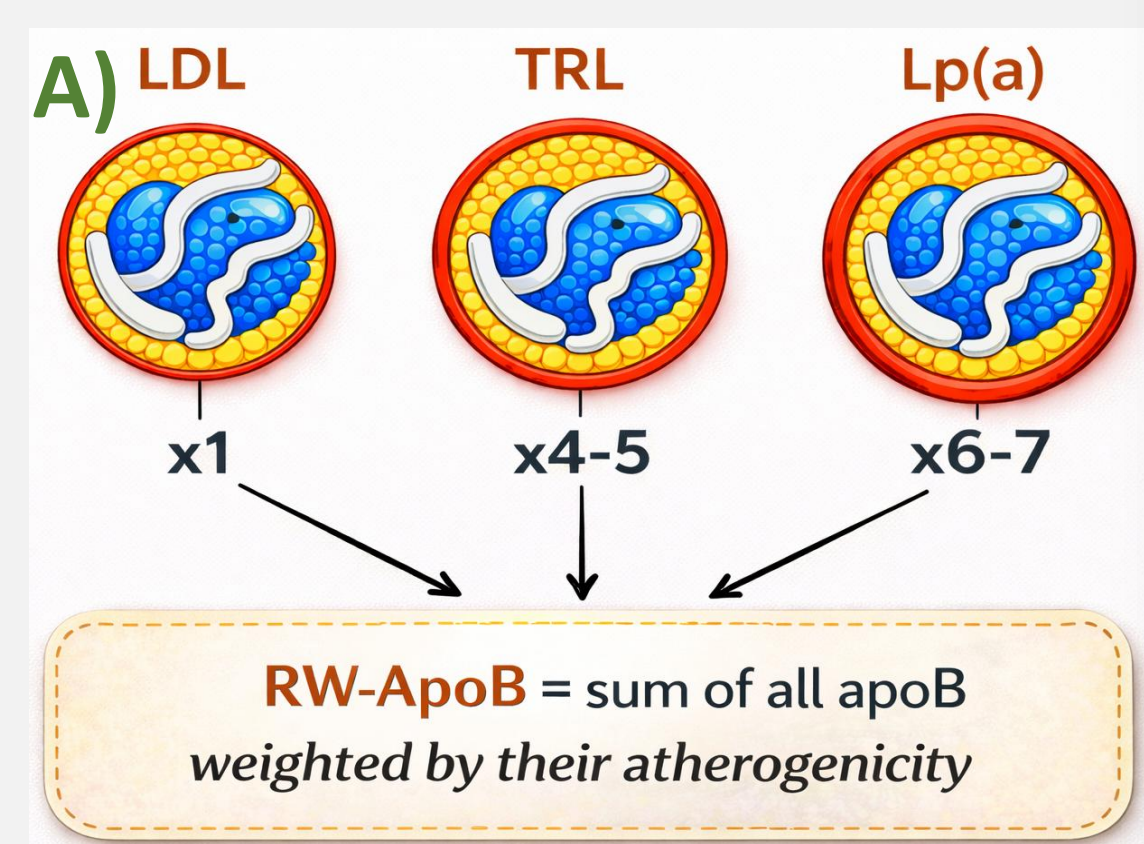


Figure 3, Rehman MB & Björnson E et al. EHJ 2026

Background and aims of REDUCE-IT analysis

Residual cardiovascular (CV) risk persists in statin-treated patients with elevated TG despite controlled LDL-C. We evaluated whether RW-apoB better predicted CV events in the placebo arm of REDUCE-IT compared to other biomarkers.

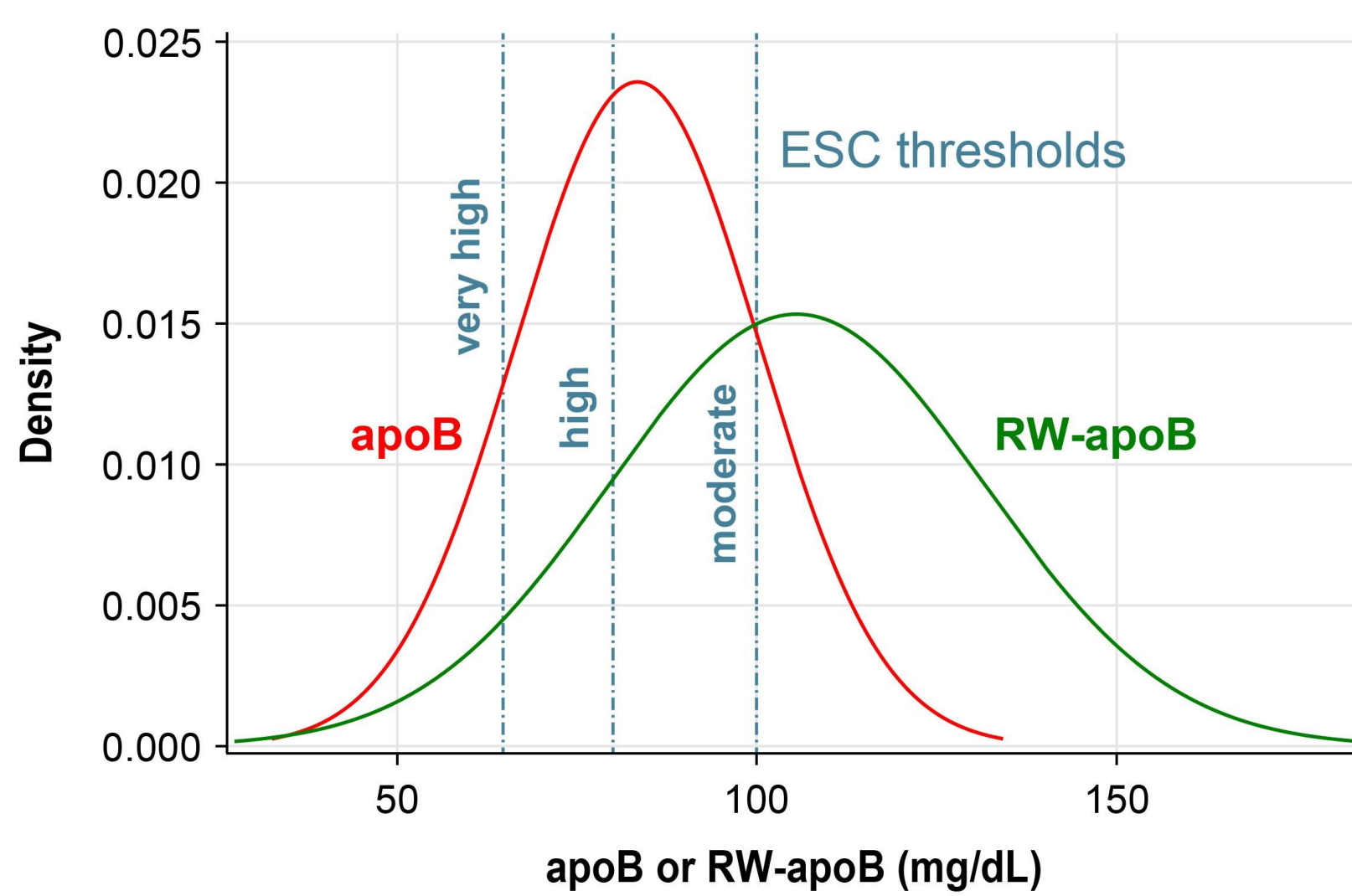
Methods

REDUCE-IT enrolled statin-treated adults with TG 135–499 mg/dL and LDL-C 41–100 mg/dL. In the placebo arm, RW-apoB in mg/dL calculated as $0.132 \times \text{TG (mg/dL)} + 0.473 \times \text{Lp(a) (mg/dL)} + 0.736 \times \text{apoB (mg/dL)}$ was determined for all 3485 participants (median age 64 years, 71% men) with these measurements available. Median follow-up was 4.9 years. We evaluated the association of lipid biomarkers with the primary endpoint (CV death, nonfatal myocardial infarction (MI), nonfatal stroke, coronary revascularization, or hospitalization for unstable angina), and with CV death/nonfatal MI. Univariable Cox models evaluated LDL-C, non-HDL-C, apoB, and RW-apoB as predictors. A multivariable Cox model included apoB + RW-apoB.

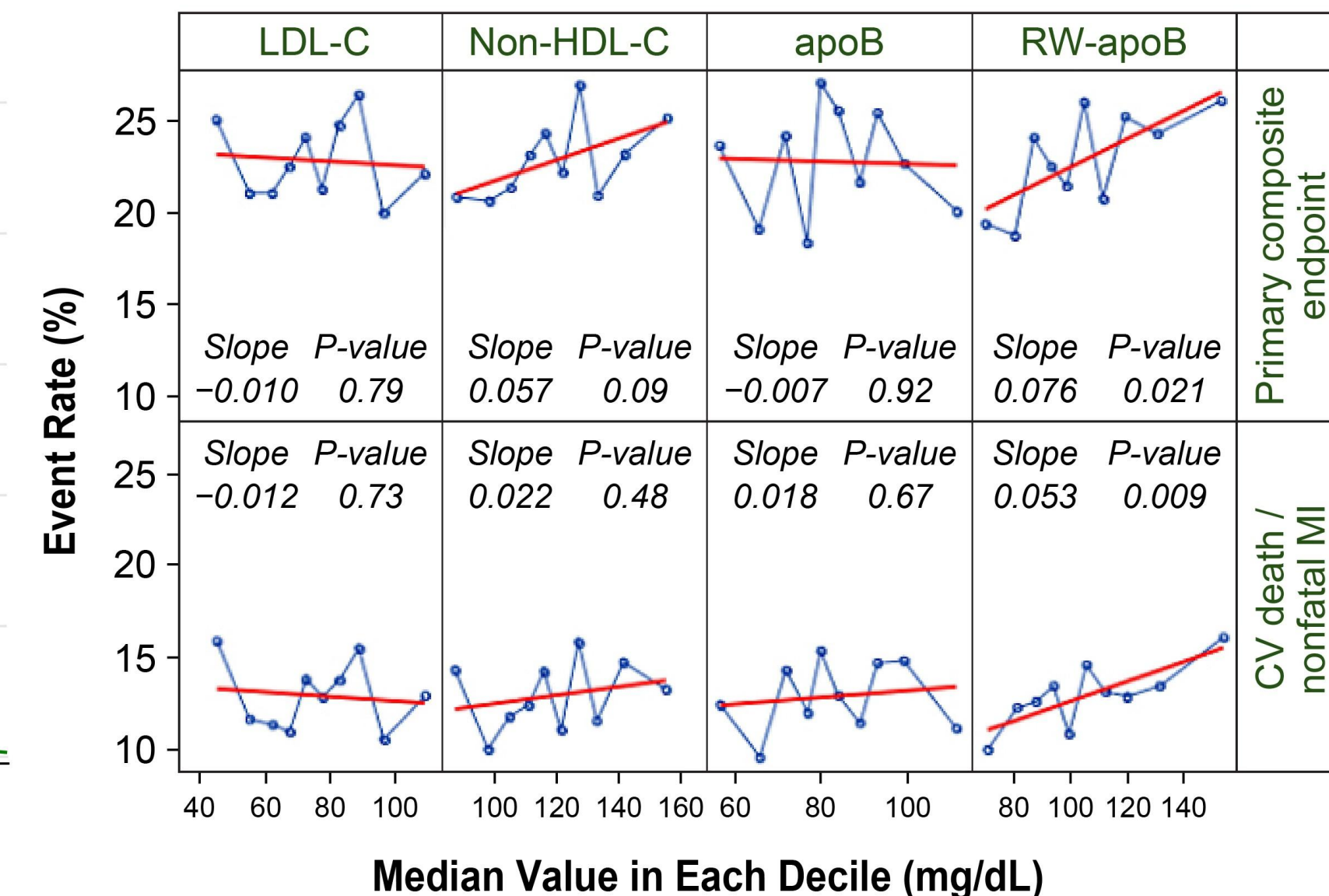
Results

RW-apoB was on average 30% higher than apoB indicating that RW-apoB may identify unrecognized lipid-associated residual risk in this group. Based on ESC intervention thresholds, more subjects were identified at risk using RW-apoB compared with apoB (Panel C).

C) apoB AND RW-apoB DISTRIBUTION



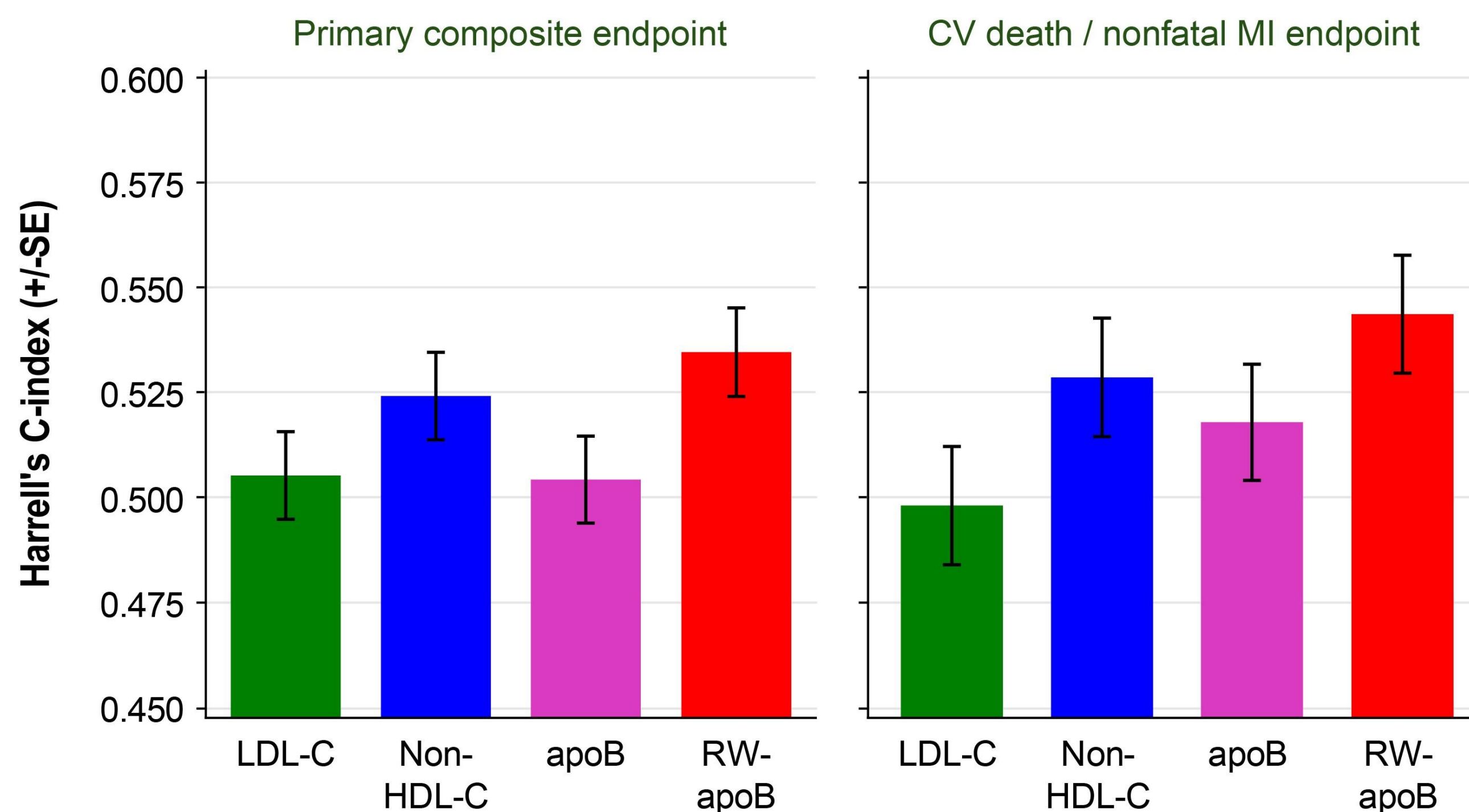
D) EVENT RATE PER DECILE LIPID BIOMARKER



RW-apoB was more strongly associated with CV risk than LDL-C, non-HDL-C, and apoB (Panel D).

E) RISK PREDICTION ACCURACY

RW-apoB had better prediction accuracy for the primary endpoint and CV death/nonfatal MI (Panel E).



Note: A Harrell's C-index of 0.5 represents no predictive capacity above chance.

In both univariable and multivariable models RW-apoB was positively associated with the primary endpoint and CV death/nonfatal MI ($P < 0.005$) while apoB was not (Panel F).

F) Cox proportional hazards models

stratified by CV risk, geographic region and ezetimibe use, adjusted by age, sex and body mass index

	Term	Hazard ratio per 10mg/dL	95% CI	P-value
* Primary Composite endpoint				
Univariable	ApoB	1.030	[0.987, 1.075]	0.18
Univariable	RW-ApoB	1.054	[1.027, 1.082]	<0.0001
Multivariable	ApoB	0.943	[0.889, 1.002]	0.06
	RW-ApoB	1.081	[1.043, 1.120]	<0.0001
Cardiovascular death / Nonfatal myocardial infarction				
Univariable	ApoB	1.042	[0.985, 1.103]	0.15
Univariable	RW-ApoB	1.056	[1.020, 1.092]	0.002
Multivariable	ApoB	0.960	[0.887, 1.040]	0.32
	RW-ApoB	1.074	[1.024, 1.127]	0.003

Conclusions: RW-apoB provided superior risk prediction in this statin-treated hypertriglyceridemic population. It better identified individuals requiring further intervention.